

Ad Hoc Semantic Reconciliation

What two reasoning agents can settle between themselves, measured across four settings and the cognition spectrum



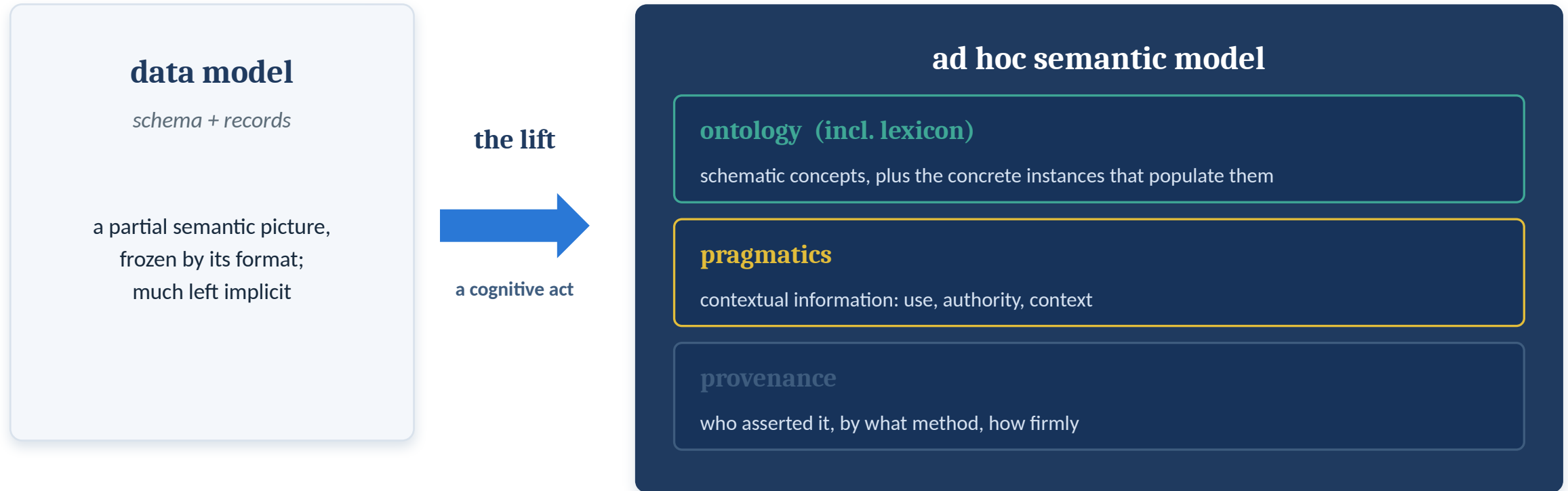
Two models, and no time to standardise

When two software systems must work together but hold different models — of the same domain, or of adjacent domains that must connect — someone has to reconcile those models. The scope a model must cover is not fixed, the useful level of abstraction depends on the task, and capable engineers, working independently, produce different but defensible models of the same network. The classical remedy is a universal standard agreed ahead of time: everyone adopts one vocabulary, and the reconciliation is done once, by committee, before the systems ever meet. That remedy is expensive, slow, and permanently behind the systems it tries to govern.

Once the systems on each side can reason, another route opens. They need not wait for a standard; they can reconcile their models ad hoc, between themselves, for the task at hand, machine to machine.

The programme puts that to the test across the cognition spectrum, from both systems reasoning to one inert to both, asking how far ad hoc reconciliation reaches at each point and what a small published reference adds. The answer is built in a measured harness, scored against a validated gold standard, rather than shown once by hand.

The lift: from a data model to a portable semantic model



A data model is already a partial semantic picture, but a fixed one, frozen by the format it uses, with much of the meaning left implicit in specifications and convention. The lift turns that frozen, partial picture into an explicit, self-describing semantic model, so any cognitive consumer can pick it up with no pre-agreed standard. Definitions, worked examples and an optional reference support the lift; they are not parts of the model.

A small family of operations

Reconciliation is not one act. Naming its operations once lets every later finding be located on a precise step of the process.

● **Lift** — data model to semantic model, per side

● **Schema binding** — align the concepts: lexical (names) and ontological (kinds, relations)

● **Instance co-reference** — decide which individual is which, sometimes by probing the live system

● **Pragmatic resolution** — settle what the reconciled thing is for, and whose authority governs it

● **Lifecycle recurrence** — re-reconcile as the live situation changes over time

● **Reference construction** — build the thin shared ground the sides bind through

● **Attribute pinning** — fix a shared field's meaning: a committed rate, not a line rate

● **Verification** — confirm a correspondence, or check that a realisation satisfies a bound

● **Composition / correlation** — group reconciled parts into wholes: several symptoms into one incident

● **Scaling** — how the total work grows with the number of systems

Where the cognition sits governs everything

Both cognitive



Reconciliation completes

Both sides reason and can be interrogated: unbounded questioning and decisive virtual experiment, so the residual is, in principle, none. This is the automatable end.

One inert



A reference buys it back

One side is a static description. Interrogation falls away; a thin published reference supplies the facts it can no longer get, up to the point of authority.

Both inert



Honest referral

Neither side can be questioned. What cannot be verified is deferred to a person rather than guessed; the residual is larger, at perfect precision.

One model ladder, one harness

THE MODEL LADDER

Three points spanning capability

 **gpt-5.6-sol**
sol · strong

 **gpt-5-mini**
mini · mid

 **gpt-5-nano**
nano · weak

Chosen so the ladder isolates model strength, not provider or architecture.

THE HARNESS

A case, scored against a gold

- A case is two lifted semantic models plus a validated gold: correct correspondences, known false cognates, and the residual expected at each placement.
- Four factors are varied: the reference (none, thin, richer), the cognition placement, the operational case, and difficulty.
- Five metrics are read: precision, resolved fraction, surviving false cognates, the residual, and reasoning tokens.

One thesis, tested four ways

1 • Configuration

ONF TAPI ↔ IETF TE / ACTN

Two public standard models of one network. The founding test: cognition reconciles them ad hoc, and a reference substitutes for it.

2 • Intent

refine • negotiate • lifecycle

A declarative intent reconciled against a concrete service by refinement. A two-sided negotiation, and the pragmatic operation enters.

3 • Cross-domain

private OSS ↔ private IP/VPN

No public standard beneath either model. The agents must build the shared ground themselves before they can bind.

4 • Observability

legacy fault mgr ↔ IETF NMOP

An alarm is not an anomaly. Meaning is settled by a standard; the whole question is significance, carried by pragmatics.

Two standard models of one network

THE CASE

ONF TAPI and IETF TEAS/ACTN describe the same optical network in different ways, each in its own vocabulary and structure. A TAPI connectivity-service is a TEAS tunnel; a TAPI link-termination-point looks like a TEAS tunnel-termination-point but is a link end, not a trail head. Both are public standards the agents recognise, which makes this the cleanest place to isolate the mechanism.

WHAT WAS PROVEN

Cognitive agents reconcile the two models correctly and ad hoc; the concept holds. Progress shows in two numbers: precision (of the correspondences an agent commits, the share that are right) and resolved fraction (of the true correspondences, the share it confirms rather than defers). The lift is the lever: matching on bare labels finds only 0.66 to 0.76 of the correspondences, and the lifted content recovers them to 0.92 to 0.97. Verification, run as its own step, rejects the false cognates, so precision climbs toward one. A thin reference then carries the strong agent to a perfect, verified close, both numbers at 1.00 across the whole spectrum, for one to two orders less deliberation. For the weaker agents the reference mainly prevents the errors they would otherwise commit, lifting hard-case precision (mini 0.88 to 0.96, nano 0.91 to 0.97) short of that perfect close. Work scales linearly with a shared reference and quadratically without, verified to $N = 12$.

Cognition completes the reconciliation; a reference partly substitutes for it, saving cognitive effort and preventing errors, and the spectrum governs both. Setting 1 does not touch pragmatics at all; that is the thread the other three settings pick up.

Refinement, negotiation, and a self-renegotiating service

THE CASE

A customer's intent (New York to Frankfurt, at least 8 Gbit/s, at most 5 ms, four-nines) is a set of bounds, not a service. An operator holds a catalogue of concrete optical services. Neither speaks the other's language; they must work out which realisation satisfies the wish and negotiate when none meets every bound.

WHAT WAS PROVEN

With both agents live, the reconciliation completes autonomously, negotiation included, at decision accuracy 1.0 for sol and mini, with no human in the loop. Remove the customer's live judgement and a pre-placed movable policy still closes the decisions it was authorised for (1.0); a mute customer with no policy cannot, and the agents correctly refer the decisions onward rather than guess. A published reference supplies information: the blind strong agent rises from 0.29 to 0.71 once given an invariant floor to check against. It does not supply authority: at both-inert no reference moves the negotiation, because what is missing is the customer's judgement. Effort climbs steeply down the ladder, from about 150 reasoning tokens for sol to 860 for mini and 5,400 for nano.

Cognition completes a negotiation. A thin published reference can stand in for the information an inert side would otherwise supply, but not for the authority to decide. A pre-placed policy is shown able to carry a party's authority to where a person or live cognitive system would otherwise have to stand.

Reconciling with no public standard

THE CASE

Meridian, a private transport OSS, thinks in circuits, bearers, wavelengths and protection grades. Cascade, a private IP/VPN controller, thinks in services, attachments, VLANs and classes of service. Their worlds touch at one seam. Both carry a grade, but Meridian's is a transport protection class and Cascade's an IP class of service: the same word, unrelated meanings, and no standard to consult.

WHAT WAS PROVEN

The central result is a pair of mirror-image failures, and neither is a success. Without a shared reference, the strong agent under-commits: it binds only the names that already coincide, resolving about half the correspondences (resolved fraction ~0.5) and deferring the rest. Its precision on what it binds is perfect, but binding rightly and far too little is still a failure. The weak agent mis-commits: it binds freely (resolved fraction 0.8 to 1.0) at a precision never above 0.83, taking the cross-domain grade trap. A thin reference remedies both - and the agents need not be handed it: run end to end, a strong agent that constructs the shared ground itself from the two models and then binds through it lifts from a no-reference resolved fraction of 0.40 to 0.93, near the reference-given close, at perfect precision with no false cognate. Stripping the reference's fields one at a time shows what it must carry: any single descriptive field unlocks the strong agent, while a bare shared identifier with no description is worse than nothing (precision 0.50). The reference works through shared description, never through the pointer.

With no public standard, cognition still closes, but only once it has built the shared ground, and building that ground is the work. The reference reaches meaning; it does not reach authority.

An alarm is not an anomaly

THE CASE

A legacy fault manager emits one alarm bundling an event, an undesirable state, a fixed severity and a probable cause. An IETF NMOP agent speaks the RFC 9940 ladder, which separates what legacy conflates and annotates each anomaly with a concern score, a confidence score, context and lifecycle. Whether a rising bit-error-rate warrants a page is not settled by the reading alone; it turns on the operational context around it, since a maintenance window, for instance, makes the same deviation expected.

WHAT WAS PROVEN

Three results. The ontological cognate, alarm against anomaly, is a clean three-rung gradient: the strong agent never conflates them; the mid agent does once a side is inert, and the RFC 9940 reference rescues it completely; the weak agent conflates them regardless. In the verdict task the pragmatics carry the operative meaning: with them on, the strong and mid agents suppress correctly during maintenance and the false-page storm disappears (accuracy 1.0 and 0.83); with them off the pipeline pages nearly everything. The payoff is power-gated: handed identical annotations, the weak agent barely moves (0.53 on, 0.50 off). Correlation is different: given the dependency map, every model, the weak one included, folds related symptoms into one incident.

Meaning and significance are separable, both necessary, and each gated in its own way. Names, definitions and references settle what a thing is; they cannot settle whether it matters or what to do about it. That layer of meaning, the pragmatics, is decisive here, and only an agent of sufficient capability can carry it.

Ad hoc reconciliation works

Two cognitive agents reconcile independently authored, divergent models correctly, with no standard agreed in advance. They complete an equivalence between two standards, negotiate an intent, bind across a private domain boundary, and judge what an observability signal means. Every case is scored against a validated gold standard across the cognition spectrum.

At the fully-cognitive end, the reconciliation completes autonomously in every setting: no standard agreed in advance, no human in the loop.

This holds even for the two operations that look least automatable: **a two-sided negotiation (decision accuracy 1.0) and a significance verdict (accuracy 1.0 with the pragmatics on).**

Everything else in the synthesis qualifies this: how automation degrades as cognition recedes, and what a thin reference buys back.

Full cognition makes a reconciliation self-verifying

Placement on the cognition spectrum decides what a reconciliation can bind. It also decides something that cuts across all four settings: whether the agents can verify their own result, confirming the binding is right rather than merely asserting it.

With full cognition on both sides, the same mechanisms that close the residual, mutual interrogation and decisive virtual experiment, also confirm the close. The agents verify their own result rather than assert it.

As a side goes inert those mechanisms fall away, verification degrades to a weaker check by satisfaction, and the assurance that the reconciliation is correct weakens with it. Full cognition does more than improve accuracy: it is what makes an ad hoc reconciliation safe to automate, with no standard and no person in the loop.

What the four settings establish

- T1** It is cognition that completes a reconciliation: descriptor methods carry it most of the way, then reach a ceiling and stop.
- T2** Where the cognition sits is the master factor: the further it recedes from both agents, the more the reconciliation leaves unresolved.
- T3** A thin reference stands in for some of the reasoning and supplies facts an inert side lacks, but never provides the authority to decide.
- T4** Strong and weak agents fail in opposite directions: the strong omit, deferring what they cannot verify; the weak commit, binding wrongly.
- T5** The pragmatic layer, what a thing means in context and whether it matters there, lies beyond names and references; it is decisive and gated by the agent's power.
- T6** A shared reference makes the work scale linearly rather than quadratically with the number of systems; which fields it carries also matters.

The four evidence readings that follow take these along four cross-cutting axes: cognitive load, the reference, placement and failure, and the pragmatic layer.

Capability buys economy as much as correctness

To reach the same decision, the weak agent spends one to two orders of magnitude more reasoning than the strong one. Effort was metered directly in the two settings that turn on a discrete decision, the intent negotiation and the observability verdict, where the reasoning to reach it can be counted; the pattern holds in both. A weak agent does not merely make more mistakes; it burns far more cognition making them.

Intent negotiation

reasoning tokens to a decision

sol ~150

mini ~860

nano ~5,400

about 36× sol to nano

Observability verdict

reasoning tokens to a decision

sol ~60

mini ~520

nano ~1,200

about 20× sol to nano

One thin reference, a different benefit at each end of the cognition spectrum

CORRECTNESS VALUE

Concentrates where cognition is weak

The weak agent is the one that commits false cognates; the reference disciplines it out of them. A strong agent was already going to get those right, so there is little correctness left to add.

EFFORT VALUE

Peaks where cognition is strong

A capable agent that would otherwise reason its own way to the answer leans on the reference instead, cutting its effort by one to two orders. In the configuration setting this collapse in the strong agent's deliberation, with the reconciliation still perfect, is the reference's main payoff.

...but its potential fields are not of equal value

Even one descriptive field (a shared label, a definition or an example) lets a capable agent complete the binding. A bare shared identifier with no description is worse than nothing (precision 0.50); a shallow class tag can even mislead the weak agent it was meant to help. A reference works through shared meaning, never through the pointer.

The further cognition recedes, the more is left unresolved

Probing resolves the hard cases until a side goes inert

instance co-reference • setting 1

- At both-cognitive, a live side can be interrogated: more probing drives the residual to zero, so the only limit is how much you probe. This is the budget-limited case.
- At one-inert, there is nothing live to probe, so no amount of probing helps and the residual sticks near 0.08. The limit is now structural.
- The same probing is run at both placements; whether it resolves the cases is set by placement, not by how much effort is spent.

Strong and weak agents fail in opposite ways

the strong omit • the weak commit

- Denied the ground it needs, a strong agent defers: it leaves the hard cases unresolved, at perfect precision on what it does bind.
- A weak agent commits: it binds freely and wrongly, taking the false cognate. It is no less confident on a wrong answer than on a right one.
- In the cross-domain setting the two break in opposite, symmetric ways on the very same case - and across six models it resolves to a smooth gradient from omission to commission, not a cliff.

Decisive for meaning and gated by cognitive power

What a reconciled thing is for, whether it matters in its context and what to do about it, lies beyond names, glosses and references. It is settled from the pragmatic information a lifted model carries: use, authority, context. The observability setting shows this by example, in two operations that resolve significance from that information:

The verdict

judgement • power-gated

Deciding whether an anomaly warrants a page needs judgement. With the pragmatics on, sol and mini suppress correctly during maintenance (1.0 and 0.83); handed the identical annotations, the weak agent barely moves (0.53 on, 0.50 off).

Correlation

structural input • robust

Folding related symptoms into one incident needs only the dependency map. Given it, every model on the ladder, the weak one included, correlates correctly; without it, every model fails.

Both draw on that pragmatic information and both are decisive; but the verdict is a judgement while correlation only applies a structural map. That difference, not the label, decides whether a weak agent can do it.

The cognition spectrum and what happens at each placement

Placement	What the reconciliation can do	The residual	What a reference buys
both-cognitive	completes autonomously in every setting: bind, negotiate, verdict	none, in principle	substitutes for cognition (effort down 1-2 orders)
one-inert	live side reconstructs the mute one; some resolutions turn structural	grows; the shortfall appears	supplies missing information; rescues the mid agent
both-inert	a third party can only propose for adjudication	largest; every judgement referred onward	supplies information, not authority

The three rows are the master variable; every finding in the synthesis locates on one of them.

Outside the spectrum: a pre-placed policy (cognition captured in a portable artefact ahead of time) can partly stand in for a live agent in some cases, closing the decisions it was authorised for and carrying that authority.

The operations, and where each was put under test

Operation	1 • Configuration	2 • Intent	3 • Cross-domain	4 • Observability
Lift	the lever (0.66→0.97)	reused	both sides private	alarm decomposed
Reference construction	given (lexical)	given (invariant)	constructed 0.40→0.93	given (RFC 9940)
Schema binding	reference substitutes	shared ground	omission vs commission	ontological cognate
Attribute pinning	—	bound vs measured	committed vs line rate	severity / scores
Instance co-reference	budget → structural	measured (endpoints)	bracketed (repro s.1)	measured; repro s.1
Verification	own step	by satisfaction	isolated pass	verdict stands in
Pragmatic resolution	deferred	movable policy	authority attribution	verdict, power-gated
Composition / correlation	—	—	—	robust (all models)
Lifecycle recurrence	—	four-hop loop	—	—
Scaling	linear vs quadratic	—	—	—

Four surprises worth stating as findings

1

A strong agent scoring lower is often behaving better

Blind at both-inert, it refuses to affirm what it cannot verify: honest deferral rather than weakness (setting 2: 0.29 against 0.71). Read the accuracy without the precision beside it and you misjudge it.

2

A bare shared pointer is worse than no reference

An opaque identifier with no description drops precision below the no-reference floor. The reference works through shared description, not the token; even a shallow class tag can mislead.

3

What lets a weak agent resolve more also gives it more to get wrong

The same lifted content that helps it find correspondences also hands it more material to misfire on. The lift is unambiguously safe for the strong agent and double-edged for the weak.

4

Two operations in one setting demand opposite things

In observability the verdict is a judgement while correlation only applies a structural map. Both resolve significance from the same pragmatic information, but only the demand for judgement predicts which a weak agent can do.

How far an agent reaches, and that the findings replicate

Reach: what an agent can see and ask

Hold the strong agent at its best and vary only what it may SEE (how much of a record) and what it may ASK of the live system.

Seeing more caps the hardest cases at one-half. Being able to ask the OPEN question - '*what facts do you have?*', not a named field - takes it to a full close, at perfect precision. The decisive reach is the question you did not know in advance to ask.

Across the ladder the same access lands by capability: the strong agent converts it to a full close; the mid plateaus; the weak posts a high resolved fraction but with slipping precision. Power sets the ceiling reach can reach.

Breadth: the findings replicate

Each setting is now exercised on THREE independently-built cases, not one - different domains, vocabularies, traps.

Every signature reappears: the strong agent closes and the reference prevents the weak agent's errors (configuration); the omit-vs-commit mirror, repaired by a reference (cross-domain); refinement and negotiation complete while the lifecycle grades with capability (intent); the alarm-is-not-an-anomaly look-alike reliably avoided (observability).

It closes the single-constructed-case worry as far as in-house work can - but not real network data (same hand built these), which is what remains.

When is building the shared reference worth the cognition it costs?

Constructing the reference is itself an act of cognition. Weighing that cost against what it buys - the close reached versus the reasoning spent - turns "always build one?" into a rule.

Load-bearing - where no standard exists

A capable agent with no reference finds only half the true matches (resolved fraction 0.50); building the reference itself carries it to 0.93, for a little extra cognition. With no standard to bind through, that construction spend is what buys the close.

Redundant - where an effective reference already exists

In configuration and observability, binding through a given reference reaches the same close for a fraction of the cognition - a strong agent binds through it in about 100 reasoning tokens against about 1,100 to construct one. Building what you already have is wasted effort.

Counterproductive - for a weak agent, anywhere

Construction is capability-gated: the weak agent's cost explodes to about 14,000 reasoning tokens and the close it reaches gets worse, not better. Never hand construction to a weak agent.

Build the reference where there is none and the agent can build a good one; otherwise bind through the one that exists - a standard is simply a constructed reference, amortised across everyone who binds through it.

A demonstration of how it works, not a statistical survey

Each setting is built from seeded cases, designed to exercise one mechanism and seed the look-alike traps the agents must avoid. To guard against any one case being chosen to work, each setting is exercised on three independently-built cases, and the signatures hold across them (previous slide). The study shows the mechanism works and how it works; it does not estimate how often it would succeed across a whole population of real networks. What remains is now genuinely external.

Larger and more varied cases

Each setting now rests on three constructed cases, but all are authored. More and larger cases, drawn from real networks rather than authored, would show how far these patterns generalise beyond the demonstrations.

Real operational data

The models, goods and traps are authored, if validated. Grounding them in the genuine divergence between fielded systems is something only carrier collaboration can supply.

Standards-body validation

The reference lexicons and the RFC-anchored modelling choices, stated openly here, would gain from review inside NMOP and the wider standards community.

IN ONE PARAGRAPH

Cognition completes the reconciliation; the pragmatic layer is where the work now lies

Two systems that can reason do not need a standard agreed in advance to work together. They lift their data into portable, self-describing semantic models and reconcile those ad hoc, for the occasion. Across configuration, intent, cross-domain and observability, at the fully-cognitive end the reconciliation completes autonomously in every case, negotiations and significance verdicts included, with no standard and no human.

A thin reference earns its place: it partly substitutes for a capable agent's reasoning and supplies the facts a live agent can no longer get from an inert side. Across many systems, it also makes the reconciliation work grow linearly instead of quadratically. But its reach ends at information; it never provides the authority to decide.

What remains is to push into that pragmatic layer and to pin down how capable an agent must be to work there.